SECTION 4 – DOMESTIC MAGNET DELIVERY

4-1 IN-TRANSIT HANDLING GUIDELINES

TABLE 4-1
LIFTING / HANDLING INFORMATION

Maximum Weight[a]	Maximum Tilt[b]	Forklift Capability
25,000 pounds (11,340 kg)	30 degrees	Yes

[a] Includes 21,700 pounds (9843 kg) for the magnet with lifting rails and 100% fill of liquid helium, 300 pounds (136 kg) for the enclosure and 3000 pounds (1225 kg) for the gradient coil and associated hardware.
[b] Maximum tilt allowed only when supported by feet for forklift / rigging operations or moving on inclines (base lift).

CAUTION

The magnet CANNOT be shipped by train as serious damage to the magnet's internal suspension system may occur due to the vibration loads introduced by rail systems.

TABLE 4-2
SHIPPING INFORMATION

Allowable Shipping Modes	Shipping Capability	Maximum Transit Time	Comments
Airplane*, air ride only trailer, boat or ocean-going ship*	Warm or Cold	Cryogen refill for cold shipment = 15 days	Only tie downs to magnet feet and lifting rails are permitted.

* Only for crated magnets. See Section 3, International (Air / Sea) Magnet Delivery.

TABLE 4-3
MAXIMUM LOAD INFORMATION

Maximum Longitudinal Load	Maximum Lateral Load	Maximum Vertical Load	Maximum Shock Loads
1.2 G	1.5 G	1.5 G	None Allowed

Note
Only tie downs to magnet feet and lifting rails are permitted.

4-2 IN-TRANSIT SERVICE

WARNING!

IN-TRANSIT SERVICE MUST BE PERFORMED BY QUALIFIED PERSONNEL ONLY AND IN STRICT CONFORMANCE WITH DIRECTION 2325852, SET-UP AND CALIBRATION, SECTION 4, LIQUID HELIUM FILL, AND 2301164PRE, *MR MAGNET - SAFETY REQUIREMENTS*.

In-transit helium refill is performed based upon magnet shipping date. The procedure MUST be performed in STRICT conformance with Direction 2325852, Set-Up and Calibration, Section 4, Liquid Helium Fill, and 2301164PRE, *MR Magnet - Safety Requirements* (included with Direction 2325852).

In-transit electrical checks are covered in Section 5 of this manual.

4-3 EQUIPMENT / TOOLS FOR MAGNET INSTALLATION

4-3-1 Forklift

TABLE 4-4
FORKLIFT REQUIREMENTS

<table>
<tr><th>Item</th><th>Qty.</th><th colspan="2">Equipment Specification / Rating</th><th>Furnished By</th><th>Intended Use</th></tr>
<tr><td rowspan="3">Forklift</td><td rowspan="3">1</td><td>Forklift Requirement:</td><td>25,000 pounds (11,340 kg) load at 50 inches (1,270 mm) to load center</td><td rowspan="3">Rigger</td><td rowspan="3">Unloading or moving magnet (without pallet)</td></tr>
<tr><td>Minimum Fork Length:</td><td>90 inches (2,286 mm)</td></tr>
<tr><td>Minimum Distance Between Forks:</td><td>80 inches (2,032 mm)</td></tr>
</table>

4-3-2 Crane Lift

TABLE 4-5
CRANE REQUIREMENTS

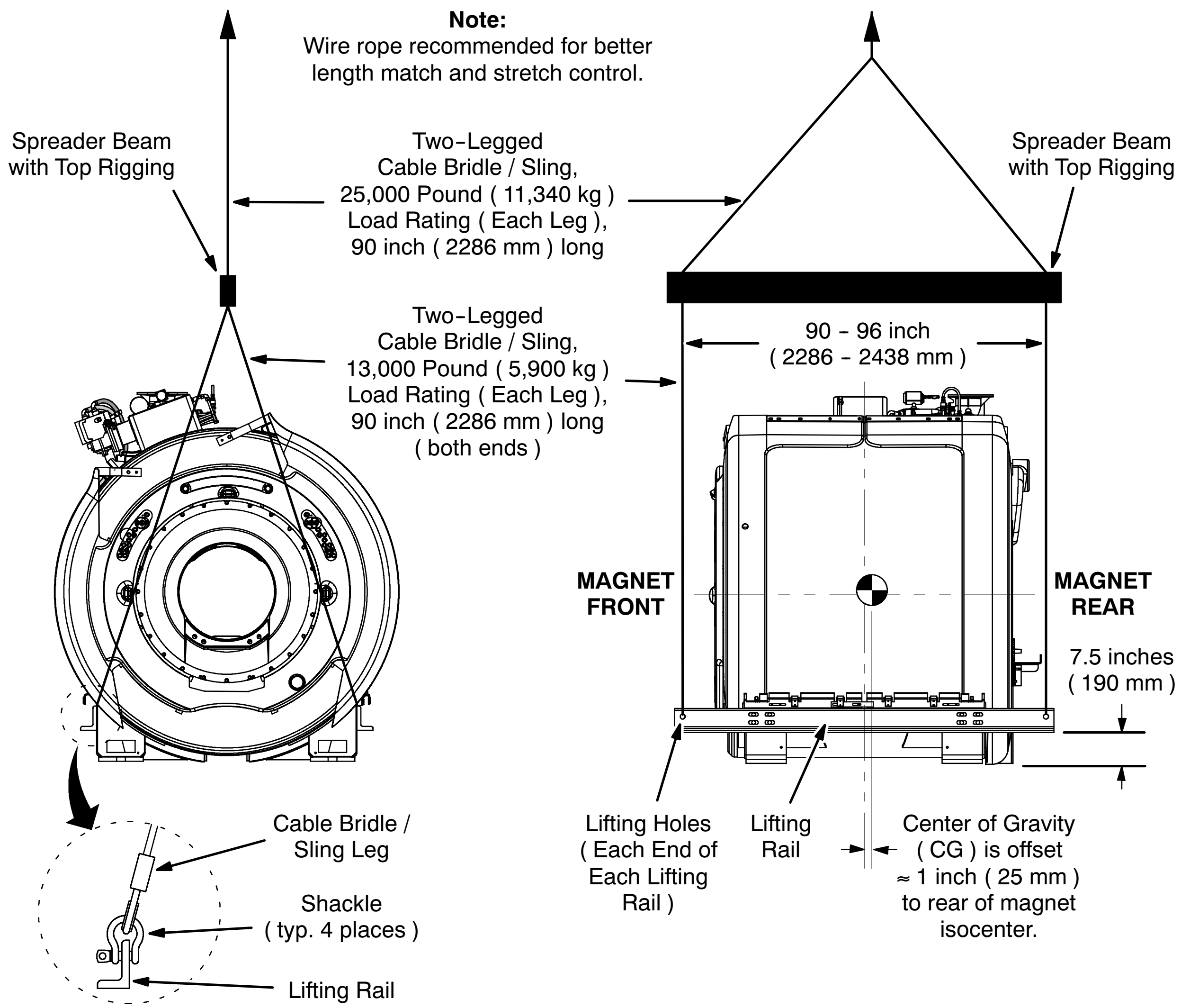
Item	Qty.	Equipment Specification / Rating		Furnished By	Intended Use
Crane	1	Total Load:	25,000 pounds (11,340 kg), minimum	Rigger	Unloading or moving magnet (without pallet)
Spreader Beam	1	Total Load:	25,000 pounds (11,340 kg), minimum	Rigger	Unloading or moving magnet (without pallet)
		Distance Between Lift Points, Under Side of Spreader Beam:	90 - 96 inches (1,219 - 1,270 mm)		
Two-Legged Cable Bridle / Sling (Lifting Rail to Spreader Beam)	2	Total Load:	25,000 pounds (11,340 kg), minimum	Rigger	Unloading or moving magnet (without pallet) Unloading or moving magnet (without pallet)
		Leg Length:	90 inches (2,286 mm) min. (all legs same length)		
		Load Per Leg:	13,000 pounds (5,900 kg), minimum		
Two-Legged Cable Bridle / Sling (Spreader Beam to Crane)	2	Leg Length:	90 inches (2,286 mm) min. (all legs same length)	Rigger	Unloading or moving magnet (without pallet) Unloading or moving magnet (without pallet)
		Load Per Leg:	25,000 pounds (11,340 kg), minimum		
Shackle	4	Pin Diameter:	1 inch (25 mm)	Rigger	Unloading or moving magnet (without pallet)
		Minimum Work Load Limit:	13,000 pounds (5,900 kg) / shackle, minimum		
Chain Hoist	2	Minimum Length:	90 inches (2,286 mm) (Actual length required to be determined by rigger.)	Rigger	Unloading or moving magnet (without pallet) through raised opening in exterior wall
		Minimum Work Load Limit:	13,000 pounds (5,900 kg) / chain hoist, minimum		

4-3-2 Crane Lift (continued)

WARNING!

MAGNET IS AN UNBALANCED LOAD. PRECAUTIONS MUST BE TAKEN TO PREVENT TILTING AND A RESULTING HAZARDOUS CONDITION:

- **MAKE SURE LIFTING APPARATUS IS IN CONFORMANCE WITH ILLUSTRATION 4-1.**
- **ORIENT SPREADER BEAM PARALLEL TO LIFTING RAILS.**
- **ADJUST LIFTING CABLES / SLINGS AND SPREADER BEAM LIFT POINT TO LEVEL MAGNET BEFORE FULLY LIFTING MAGNET OFF SURFACE.**



CRANE LIFT CONFIGURATION
ILLUSTRATION 4-1

4-3-3 Miscellaneous Equipment / Tools

TABLE 4-6
MISCELLANEOUS EQUIPMENT / TOOLS

Item	Qty.	Equipment / Tool Requirement	Responsible	Function
Hydraulic or Toe Jack	4	Must support one end of magnet on two jacks or both ends of magnet on 4 jacks. Total weight: 25,000 pounds (11,340 kg), minimum	Rigger	Raise magnet for roller dollies or leveling plates
Roller Dollies	4	Total weight for all dollies combined: 25,000 pounds (11,340 kg), minimum	Rigger	Moving magnet to magnet room
Level	1	24 - 36 inch (610 - 915 mm) length	Rigger	Level the magnet
Magnet Leveling Kit	1	N / A	GE / SYS	46-260888G4 Level the magnet
Magnet Boltdown / Seismic Kit	1	Apply sufficient torque to provide 2,500 pounds (1,134 Kg) clamping force per anchor. Torque requirements to be supplied by screen room vendor.	Screen Room Vendor	Bolt down magnet
Torque Wrench and Socket	1	Torque range: 58 to 258 ft-lbs. Deep socket, size dependant on anchor selection	Rigger	Bolt down magnet
Socket Wrench and Sockets	1	3/4 inch socket (add sockets for crate removal)	Rigger	Remove 1/2 inch crate lag screws
Magnet Interface Drawing	1	N / A	GE / ISS	2350040IDW Identify magnet dimensions & features
Pre-Delivery Information Package	1	N / A	GE	Aid in magnet installation
Portable Temperature Meter (2125073)	1	N / A	GE / FE	Check internal temperatures
Digital Volt / Ohm Meter	1	Fluke model or equivalent	GE / FE	Perform magnet electrical checks
Magnet System Components Checks: Tables 5-1 and 5-2	1	N / A	GE / SYS	Confirm electrical check values
Wood Timber	2	4 inch x 6 inch x 8 foot (100 mm x 150 mm x 2.5 M)	Rigger	Moving / positioning magnet to magnet room
Steel Plate	6	18 inch x 6 foot, .50 inch min. thickness (500 mm x 2 M, 6.5 mm min. thickness)	Rigger	Moving magnet to magnet room

4-3-3 Miscellaneous Equipment / Tools (continued)

TABLE 4-6 (CONTINUED)
MISCELLANEOUS EQUIPMENT / TOOLS

Item	Qty.	Equipment / Tool Requirement	Responsible	Function
Pry-Bar	2	Length as required to move magnet	Rigger	Moving / positioning magnet in magnet room
RuO Temperature Meter (2171219)	1	N / A	GE	Check second stage & recondenser temperatures
Adapter Cable for Diode Temperature Meter	1	N / A	GE	Check internal temperatures
Circular or Chain Saw	AR	N / A	Rigger	For cutting wood timbers
Heavy Duty Drill & Drill Bits	AR	N / A	Rigger	To add holes to wood timbers
Min. .50 inch Dia. Mounting Hardware	AR	N / A	Rigger	For attaching wood timbers to rails

4-4 CRATE AND PROTECTIVE PACKAGING REMOVAL

CAUTION

Care must be taken not to scrape or hit the sides of the magnet.

The magnet is shipped with bubble-wrap plastic wrap and a separate outer plastic bag. The bubble-wrap should be left intact until the magnet is anchored in the magnet room. Care must be taken in removing the plastic bag to avoid damage to various magnet components.

1. The crate can be removed with a crane either by lifting straight up or by removing the bolted boards on one end and pulling the crate toward that end.
2. The outer plastic bag should then be removed.

MAGNET AS SHIPPED (PLASTIC BAG AND BUBBLE-WRAP)
ILLUSTRATION 4-2

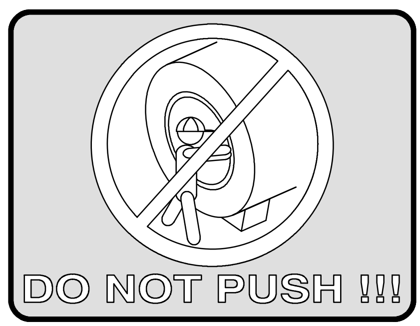
4-5 UNLOADING THE MAGNET

WARNING!

TO PREVENT PERSONAL INJURY OR MAGNET DAMAGE:

- **DO NOT APPLY ANY FORCE TO THE MAGNET'S FIBERGLASS TOP, SIDE OR END COVERS. SEE ILLUSTRATION 4-3.**
- **USE LIFTING CABLES / SLINGS AS SHOWN TO LIFT MAGNET.**
- **BEFORE MOVING THE MAGNET TO THE MR SUITE, REFER TO SECTION 5.**

Located on Both Sides



Located on Both Ends

MAGNET "DO NOT PUSH" SIGN LOCATIONS
ILLUSTRATION 4-3

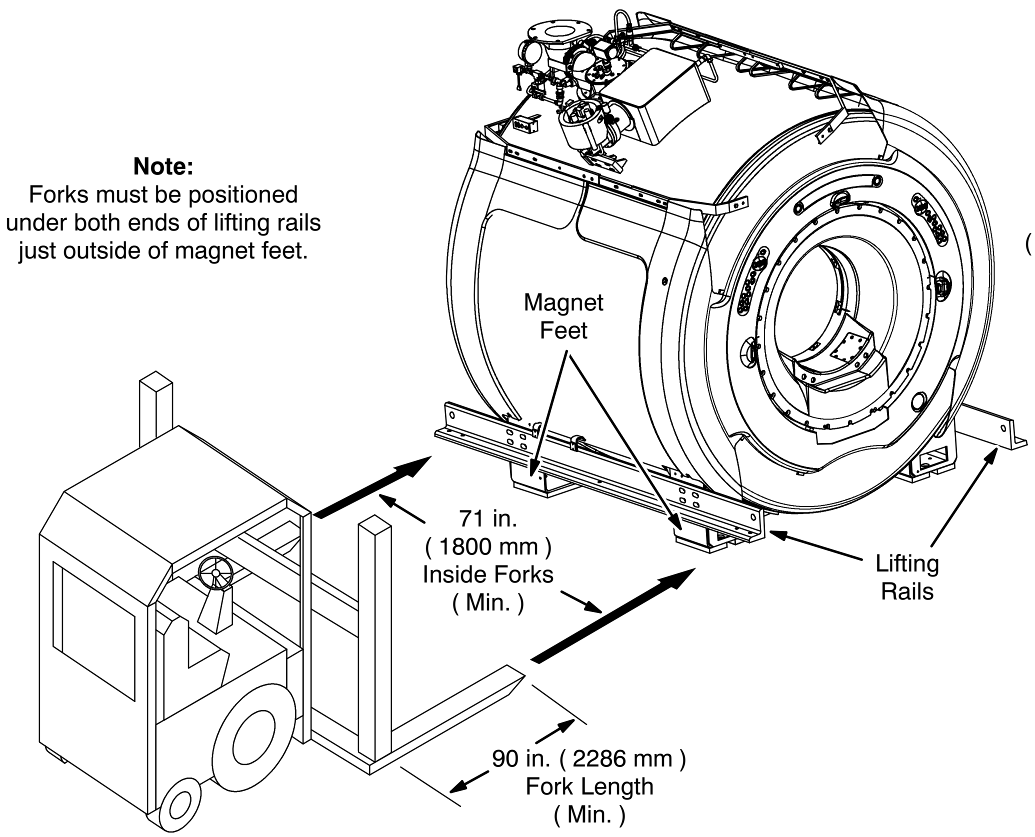
4-5-1 Forklift Unloading

CAUTION

Extreme care must be exercised during forklift operations:

- **Use the PROPER size forklift. Make sure the forklift meets the minimum capacity and dimension requirements stated in Table 4–4.**
- **The magnet MUST be picked up from the sides only. The forks must be placed under the lifting rails as indicated in Illustration 4–4.**
- **The magnet MUST be lifted smoothly to avoid impact or jolts to the system, which may cause damage to the magnet.**

1. Position the forklift at the side of the magnet and facing the magnet. Locate the forks under the lifting rails as indicated in Illustration 4-4.



FORKLIFTING UNDER LIFTING RAILS
ILLUSTRATION 4-4

4-5-1 Forklift Unloading (continued)

CAUTION

Forklift forks can damage the magnet enclosure. Use protective padding around the forks.

2. Wrap the full length of each fork with protective padding material to prevent damage to the magnet's enclosure.
3. Carefully drive the forklift until the forks are completely under both lifting rails in the areas shown.
4. Lift the forks to right below the lifting rails, adjust the distance between forks so that padded forks lightly touch the enclosure and then finish raising the forks to the lifting rails.

CAUTION

Impacts / jolts to the magnet while lifting / moving / lowering the magnet can cause expensive internal magnet damage. Lift / move / lower smoothly. Do not allow the magnet to bump or hit anything forcefully.

Avoid excessive tilt (as specified in Table 4-1) as magnet damage may result.

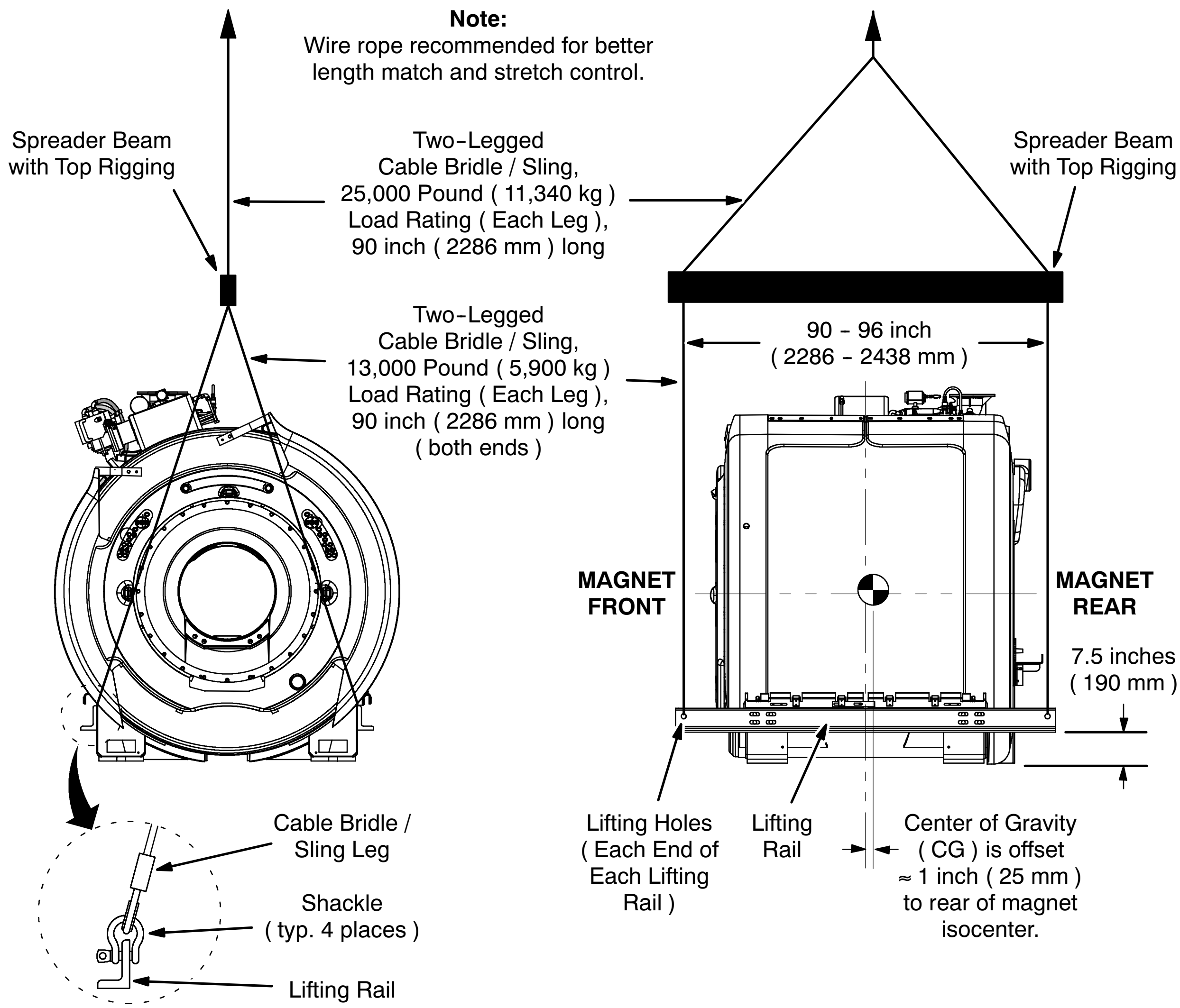
5. Lift the magnet with the forklift.
6. Smoothly move the magnet to the desired location and carefully lower to rest on a flat surface.

4-5-2 Crane Unloading

WARNING!

MAGNET IS AN UNBALANCED LOAD. PRECAUTIONS MUST BE TAKEN TO PREVENT TILTING AND A RESULTING HAZARDOUS CONDITION:

- **MAKE SURE LIFTING APPARATUS IS IN CONFORMANCE WITH ILLUSTRATION 4-5.**
- **ORIENT SPREADER BEAM PARALLEL TO LIFTING RAILS.**
- **ADJUST LIFTING CABLES / SLINGS AND SPREADER BEAM LIFT POINT TO LEVEL MAGNET BEFORE FULLY LIFTING MAGNET OFF SURFACE.**



CRANE LIFT CONFIGURATION DURING UNLOADING
ILLUSTRATION 4-5

4-5-2 Crane Unloading (continued)

CAUTION

The lifting hooks of the spreader beam MUST be located as shown in Illustration 4-5. The magnet can be lifted by a crane which meets the specifications listed in Table 4-5. Failure to follow these requirements can result in serious damage to the magnet.

1. Position the hook of a crane meeting the specifications stated in Table 4-5 centrally over the magnet to ensure a vertical lifting force on the lifting cables / slings. See Illustrations 4-5 and 4-6.

2. Attach the rigging to the lifting cables / slings at each end of the magnet. See Illustrations 4-5 and 4-6.

EXAMPLE CRANE LIFT CONFIGURATION
ILLUSTRATION 4-6

4-5-2 Crane Unloading (continued)

CAUTION

Impacts / jolts to the magnet while lifting / moving / lowering magnet can cause expensive internal magnet damage. Lift / move / lower smoothly. Do not allow the magnet to bump or hit anything forcefully.

3. Lift the magnet with the crane.
4. Smoothly move the magnet to the desired location and carefully lower to rest on a flat surface.

4-5-3 Crane Lift Through Raised Opening in Exterior Wall

WARNING!

SERIOUS INJURIES AND MAGNET / EQUIPMENT DAMAGE ARE POSSIBLE WHEN MOVING MAGNET THROUGH RAISED OPENING IN AN EXTERIOR WALL. DO NOT BEGIN UNTIL:

- **THE ACTUALLY PROCEDURE TO BE USED IS THOUGHT OUT IN DETAIL.**
- **ALL NECESSARY EQUIPMENT IS ON SITE AND HAS BEEN INSPECTED FOR SAFETY AND WEIGHT RATINGS.**
- **ALL NECESSARY PERSONNEL ARE TRAINED AND READY.**

IMPORTANT !!!

Rigger is responsible for actual equipment / procedure used to lift and move a magnet through a raised opening in an exterior wall. The following procedure only outlines the concept of one method.

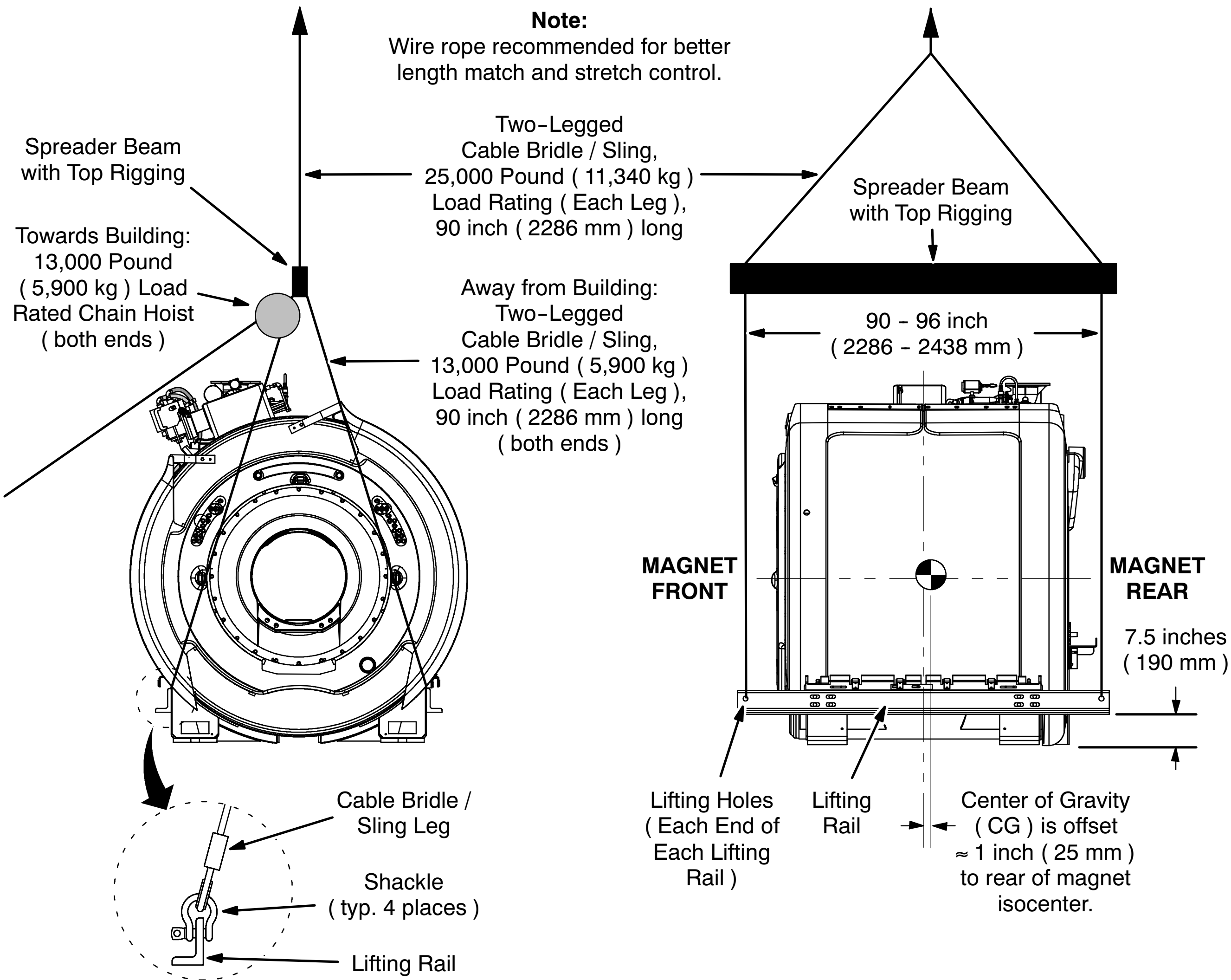
1. Make sure the opening is at least 96 inches (2440 mm) wide and 102 inches (2590 mm) tall, minimum. The magnet should pass through the opening side-first without hitting the opening. A larger opening will both make the operation easier and make accidental magnet damage less likely.

4-5-3 Crane Lift Through Raised Opening in Exterior Wall (continued)

WARNING!

MAGNET IS AN UNBALANCED LOAD. PRECAUTIONS MUST BE TAKEN TO PREVENT TILTING AND A RESULTING HAZARDOUS CONDITION:

- **MAKE SURE LIFTING APPARATUS IS IN CONFORMANCE WITH ILLUSTRATION 4-7.**
- **ORIENT SPREADER BEAM PARALLEL TO LIFTING RAILS.**
- **ADJUST LIFTING CABLES / SLINGS AND SPREADER BEAM LIFT POINT TO LEVEL MAGNET BEFORE FULLY LIFTING MAGNET OFF SURFACE.**



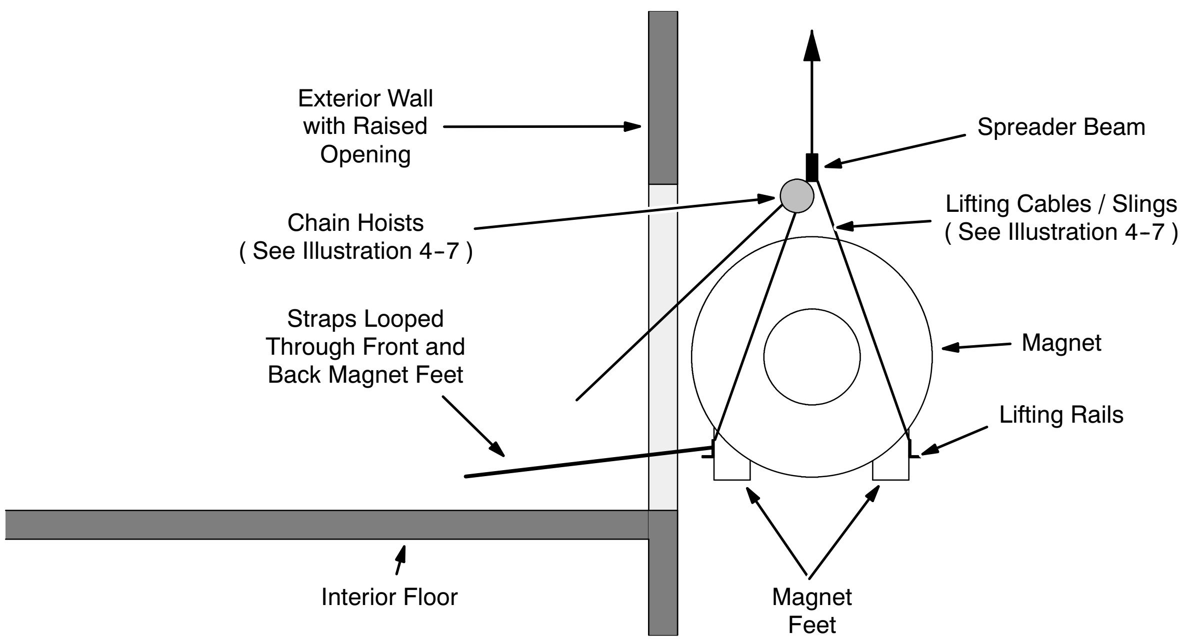
CONFIGURATION FOR CRANE LIFT THROUGH RAISED OPENING
ILLUSTRATION 4-7

4-5-3 Crane Lift Through Raised Opening in Exterior Wall (continued)

CAUTION

The lifting hooks of the spreader beam MUST be located as shown in Illustration 4-5. The magnet can be lifted by a crane which meets the specifications listed in Table 4-5. Failure to follow these requirements can result in serious damage to the magnet.

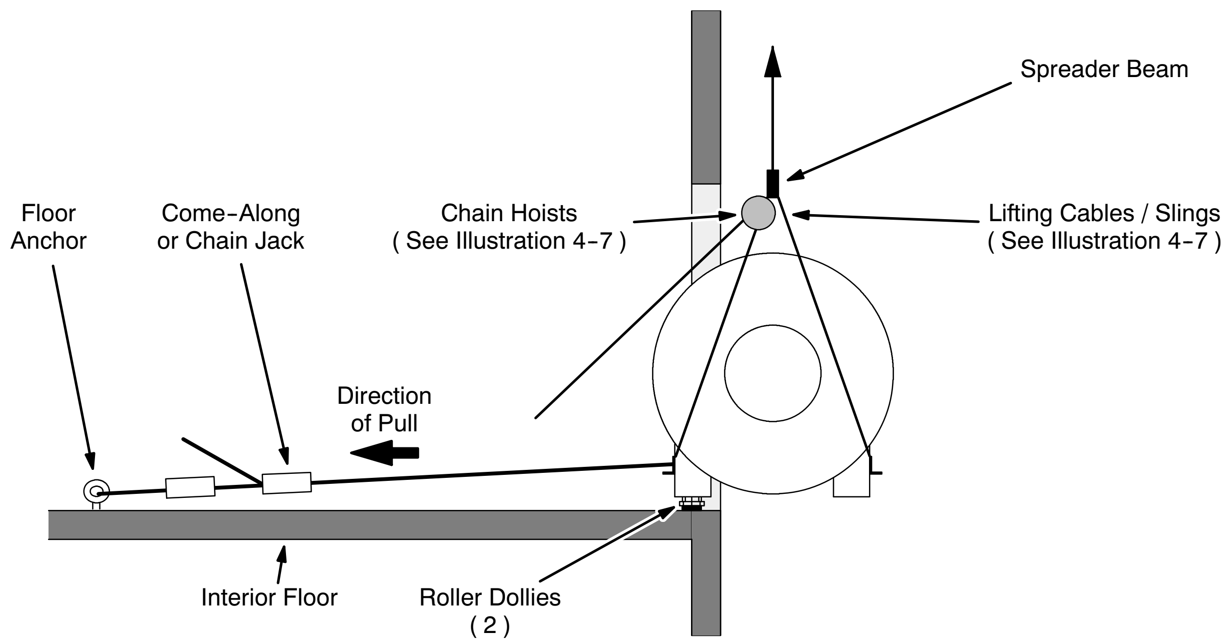
2. Position the hook of a crane meeting the specifications stated in Table 4-5 centrally over the magnet to ensure a vertical lifting force on the lifting cables / slings. See Illustrations 4-7 and 4-6.
3. Rig the magnet with chain hoists towards the building, lifting cables / slings away from the building and a spreader beam as in Illustration 4-7.
4. Attach lifting straps to both ends of the lifting rail to face towards the building. See Figure 4-8.
5. Crane lift the magnet to the height of the opening. Use the lifting cables / slings and chain hoist to guide the magnet. See Illustration 4-8.



LIFT TO RAISED OPENING
ILLUSTRATION 4-8

4-5-3 Crane Lift Through Raised Opening in Exterior Wall (continued)

6. Pull the magnet through the opening far enough to place roller dollies under the front and back magnet legs as in Illustration 4-9.



STARTING MAGNET THROUGH RAISED OPENING
ILLUSTRATION 4-9

7. Connect straps between the magnet feet and a come-along or chain jack as in Illustration 4-9.

WARNING!

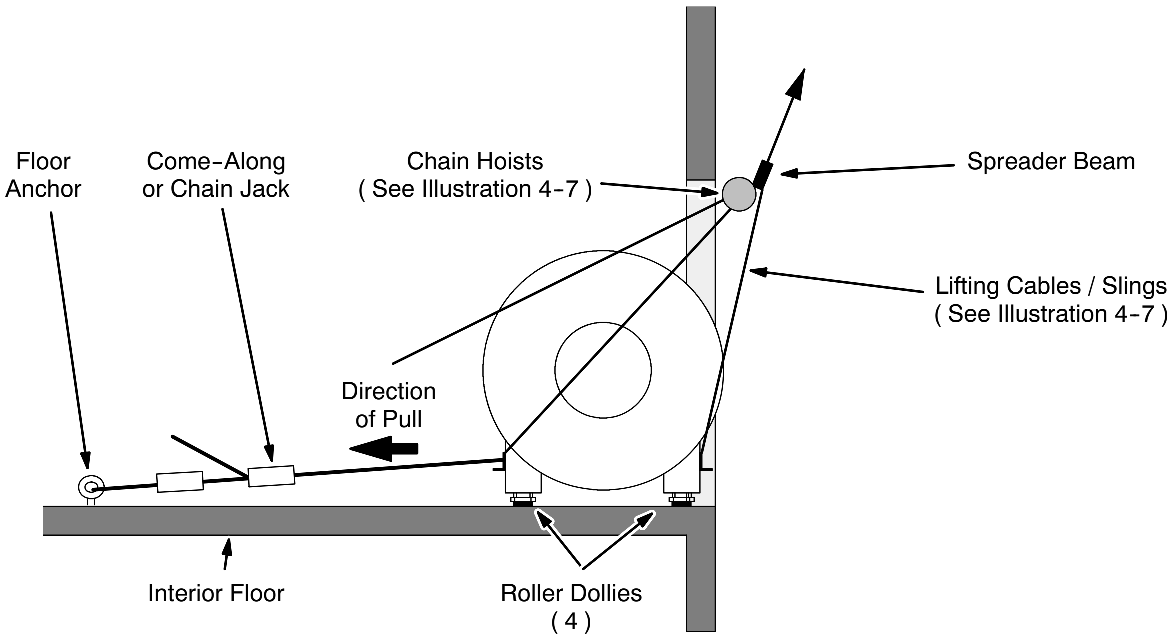
SERIOUS INJURIES AND MAGNET / EQUIPMENT DAMAGE ARE POSSIBLE IF CRANE AND CHAIN HOIST OPERATION ARE NOT CAREFULLY COORDINATED.

8. Begin pulling the magnet through the opening using the come-along while:
   - The chain hoist operator(s) carefully lengthen the chain hoist connections to the magnet in small and easy increments.
   - The crane operator raises the lifting cable connections to the magnet a matching amount.

Note
Make immediate but careful adjustments to bring magnet back to level whenever magnet starts becoming unlevel.

4-5-3 Crane Lift Through Raised Opening in Exterior Wall (continued)

9. Place a second roller dolly under the magnet's other legs when these are fully over the floor. See Illustration 4-10.



PLACING SECOND PAIR OF ROLLER DOLLIES
ILLUSTRATION 4-10

CAUTION

Do not disconnect the chain hoists or the lifting cables / slings until the magnet is completely and safely on the floor.

10. When the magnet is safely on the floor, detach the chain hoists and lifting cables / slings.

SECTION 5
MAGNET SYSTEM CHECKS
(BY GE SERVICE)

5-1 PHYSICAL INSPECTION

Note

Shipment of magnet system components to the installation site may occur as a complete system or as a drop shipment of major system components. Verify that all required magnet system components are present at the site to assure a complete and timely installation.

1. Locate the Pre-Delivery Information Package shipped with the magnet. It contains the Bill of Material for the magnet system delivered. Check that all boxes indicated are present.
2. Check the contents of each box against its packing list when the boxes are brought into the MR site.
3. Inspect the magnet for physical damage and icing / condensation on the body. If no problem is found, unload the magnet.

Note

Because of the higher boil-off and helium gas flow through the Vertical Penetration on international shipments, some frost on the Vertical Penetration may be normal during periods when the Coldhead has been shut off. Domestic shipments should have no frost.

4. If icing or condensation are present on the body or the bore of the magnet, check the liquid helium level before unloading. See Direction 2325852, SET-UP AND CALIBRATION, Section 1-6, for Cryogen Monitor set up. The occurrence of icing needs to be understood before filling the magnet with helium.

Note

If the magnet has been sitting for a period of time with the Coldhead inoperative, the magnet may be depleted of cryogens. If so, the magnet can be filled when magnet gets bolted down in the magnet room.

5-2 ELECTRICAL CHECKS

Note

It is important to establish if any damage was sustained by the magnet or its system components during delivery or if any components are missing. This will result in the fast, proper follow-up of any problems and a timely installation.

1. Make sure the Shim Lead is engaged. If needed, re-engage the Shim Lead in conformance with Direction 2192624, SET-UP AND CALIBRATION, Section 6.

5-2 ELECTRICAL CHECKS (continued)

2. Locate the Connector Pins using Table 5-1, Illustration 5-1 and Direction 2325852, SCHEMATICS / INTERCONNECTS.

3. With a digital meter measure the resistances across the identified Connector Pins and record them in Table 5-1, comparing them to the "Acceptable Range" values.

TABLE 5-1
MAGNET CIRCUITS RESISTANCE CHECK COLD (4.2K)

Function	Connector	Pin #	Resistance (Ohms)		Connector		Pin #	Resistance (Ohms)	
			Typical	Measured				Typical	Measured
Main Coil	Main Coil Power Lugs or J5-1	+ − 9, 10	< 6 Ohms						
Superconducting Shim Coils	Cannon (P1-A) on Shim Lead				Pigtail Connectors P2 & P2W				
Z1		1, 19	0.2 – 0.6		P2		A, (K, L)	0.2 – 0.7	
Z2		2, 20	0.2 – 0.6			P2W	A, (K, L)	0.2 – 0.7	
Z3		3, 21	0.2 – 0.6		P2		B, (M, N)	0.2 – 0.7	
Z4		4, 22	0.2 – 0.6			P2W	B, (M, N)	0.2 – 0.7	
C11 +		16, 19	0.2 – 0.6		P2		F, (K, L)	0.2 – 0.7	
C11 −		17, 20	0.2 – 0.6			P2W	F, (K, L)	0.2 – 0.7	
C22 +		14, 21	0.2 – 0.6		P2		G, (M, N)	0.2 – 0.7	
C22 −		15, 22	0.2 – 0.6			P2W	G, (M, N)	0.2 – 0.7	
S11 +		9, 19	0.2 – 0.6		P2		H, (K, L)	0.2 – 0.7	
S11 −		10, 20	0.2 – 0.6			P2W	H, (K, L)	0.2 – 0.7	
S22 +		7, 21	0.2 – 0.6		P2		J, (M, N)	0.2 – 0.7	
S22 −		8, 22	0.2 – 0.6			P2W	J, (M, N)	0.2 – 0.7	
C31		13, 23	0.2 – 0.6		P2		D, P	0.2 – 0.7	
S31		11, 23	0.2 – 0.6		P2		E, P	0.2 – 0.7	
Superconducting Switch Heaters Main Switch	J5-1 & J5-2 on Shim Lead (MS1-A3, A1)	1, 2	22 – 27						
Axial Shims		5, 6	7 – 9						
Transverse 1		7, 6	7 – 9						
Transverse 2		8, 6	7 – 9						

5-2 ELECTRICAL CHECKS (continued)

4. Perform the checks identified in Table 5-2. If any problems are found, the Shim Lead can be retracted to isolate shorts from the Shim Coils.

TABLE 5-2
SHIM LEAD CHECKS

P1A Connector Pins on Shim Lead	Pigtail Connectors*	Acceptable Range	Measured
Lead To Lead Shorts			
19 to 20	P2 – (K,L) to P2W – (K,L)*	> 16K Ohms	
19 to 21	P2 – (K,L) to P2 – (M,N)	> 16K Ohms	
19 to 22	P2 – (K,L) to P2W – (M,N)	> 16K Ohms	
19 to 23	P2 – (K,L) to P2 – (P)	> 16K Ohms	
19 to 24	P2 – (K,L) to P2W – (P)	> 16K Ohms	
20 to 21	P2W – (K,L) to P2 – (M,N)	> 16K Ohms	
20 to 22	P2W – (K,L) to P2W – (M,N)	> 16K Ohms	
20 to 23	P2W – (K,L) to P2 – (P)	> 16K Ohms	
20 to 24	P2W – (K,L) to P2W – (P)	> 16K Ohms	
21 to 22	P2 – (M,N) to P2W – (M,N)	> 16K Ohms	
21 to 23	P2 – (M,N) to P2 – (P)	> 16K Ohms	
21 to 24	P2 – (M,N) to P2W – (P)	> 16K Ohms	
22 to 23	P2W – (N,M) to P2 – (P)	> 16K Ohms	
22 to 24	P2W – (N,M) to P2W – (P)	> 16K Ohms	
23 to 24	P2 – (P) to P2W – (P)	> 16K Ohms	
Lead to Ground Shorts			
19 to GROUND**	P2 – (K,L) to GROUND	> 16K Ohms	
20 to GROUND**	P2W (K,L) to GROUND	> 16K Ohms	
21 to GROUND**	P2 (M,N) to GROUND	> 16K Ohms	
22 to GROUND**	P2W (M,N) to GROUND	> 16K Ohms	
23 to GROUND**	P2 (P) to GROUND	> 16K Ohms	
24 to GROUND**	P2W (P) to GROUND	> 16K Ohms	

* Paired pins inside parenthesis are interconnected inside the pigtail connector. Thusly "P2 – (K,L) to P2W – (K,L)" means "Connector P2, pins K and/or L, to Connector P2W, pins K and/or L."

** Ground defined as Shim Lead Cover.

Note

See Direction 2325852, SCHEMATICS / INTERCONNECTS, Section 2, Illustration 2-4, Wiring Diagram, S/C Shim Coil Cable 46-260726G1.

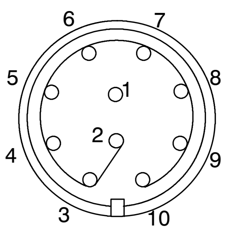
5-2 ELECTRICAL CHECKS (continued)

5. Report any damage found in compliance with the “Damage In Transportation” note on the back side of this manual’s title page.

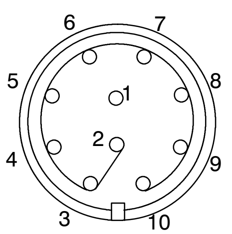
6. Report all problems found to the regional Magnet & Cryogenics (MAC) Team Leader. Report all missing components to the person identified on the magnet components Bill of Material.

Note

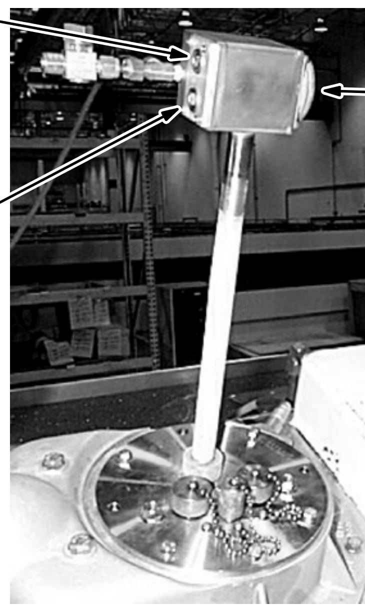
During normal operation the Shim Lead Assembly will be in the disengaged position, rotated counterclockwise until assembly locks into position.

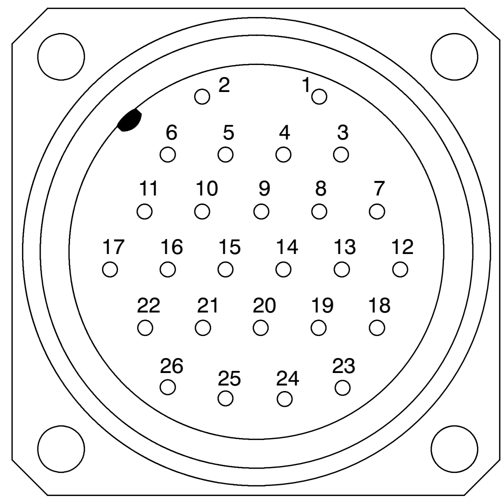


J5-1
Wiring View



J5-2
Wiring View





P1-A

Note
Connector views at Shim Lead Connector Box.

CONNECTOR PIN LOCATIONS
ILLUSTRATION 5-1

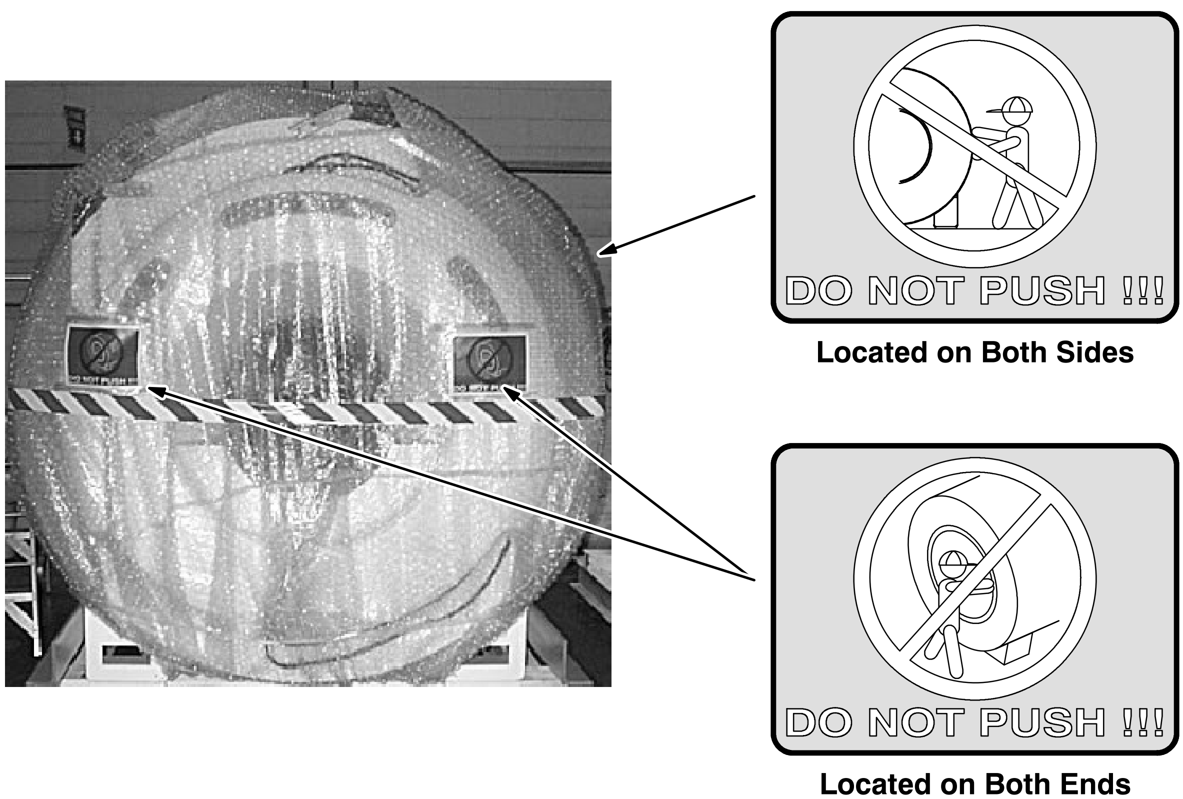
SECTION 6 – MOVING MAGNET TO MR SUITE

6–1 INTRODUCTION

WARNING!

TO PREVENT PERSONAL INJURY OR MAGNET DAMAGE:

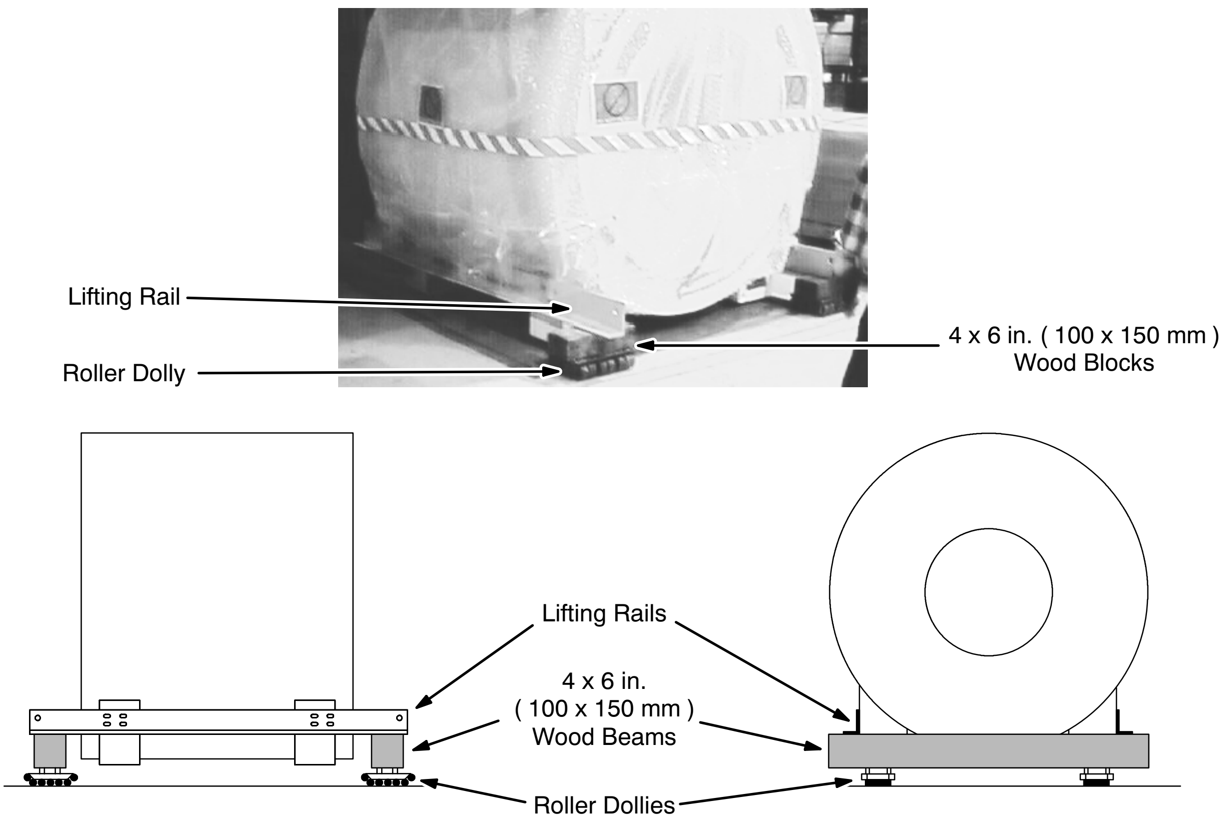
- **DO NOT APPLY ANY FORCE TO THE MAGNET'S FIBERGLASS TOP, SIDE OR END COVERS. SEE ILLUSTRATION 6–1.**
- **REFER TO SECTION 4, DOMESTIC MAGNET DELIVERY, BEFORE MOVING THE MAGNET USING FORKLIFT OR CRANE.**



MAGNET "DO NOT PUSH" SIGN LOCATIONS
ILLUSTRATION 6–1

6-1 INTRODUCTION (continued)

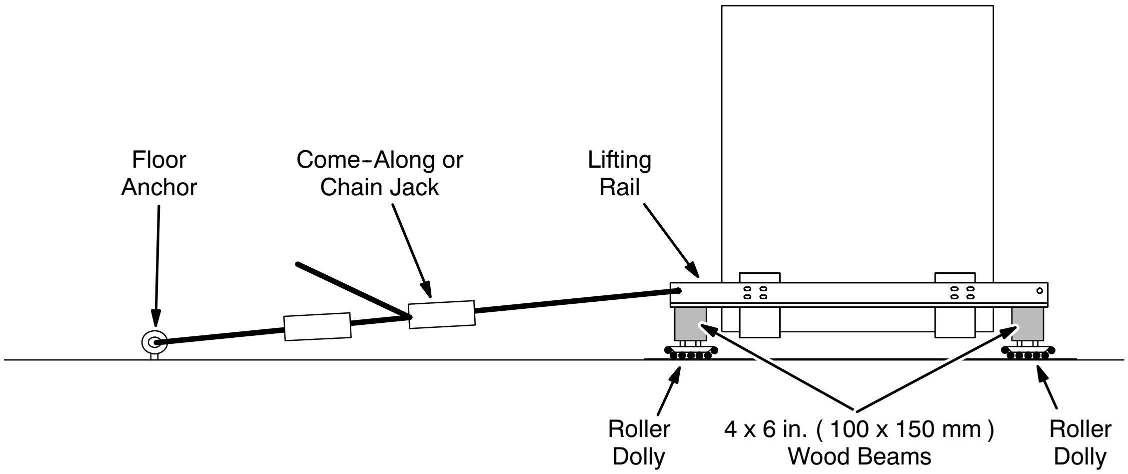
Once the magnet has been moved to the building using crane or forklift — covered in Section 4, Domestic Magnet Delivery — it needs to be moved to the magnet room. Roller dollies in the arrangement shown in Illustration 6-2 are recommended for moving the magnet inside a building. Place steel floor plates along the magnet delivery route when using roller dollies.



MAGNET ON ROLLER DOLLIES
ILLUSTRATION 6-2

6-1 INTRODUCTION (continued)

A motorized tow vehicle is recommended to help move the magnet. A come-along or chain jack such as shown in Illustration 6-3 can also be used. Attach cables, chains or straps to the lifting rails with shackles.



USING A COME-ALONG OR CHAIN JACK
ILLUSTRATION 6-3

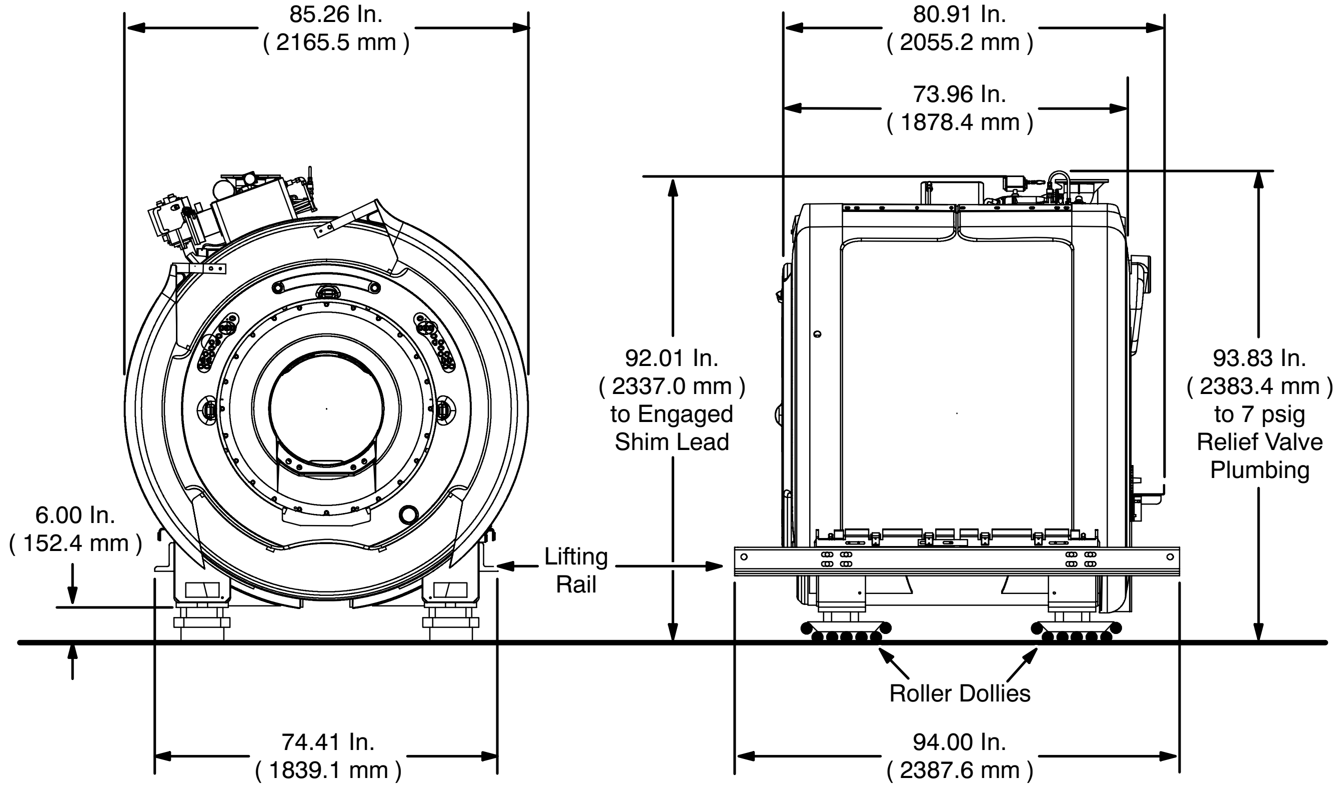
6-2 MOVING THE MAGNET

1. Identify the magnet location in the magnet room. Use tape to mark the four corners of the magnet on the magnet room floor.

Note
Refer to architectural drawings to determine the exact location of the magnet within the magnet room.

2. Check all clearances along the route the magnet will move to the magnet room. Compare those clearances with Illustration 6-4.

6-2 MOVING THE MAGNET (continued)



Notes:
- Minimum height for service clearance is 105 inches (2667 mm). This clearance is needed for Ramp Lead, Shim Lead, and fill line installation.
- Dimensions come from 2350040 rev. B, LCC300 Mechanical Interface.

MAGNET WITH ROLLER DOLLIES ATTACHED
ILLUSTRATION 6-4

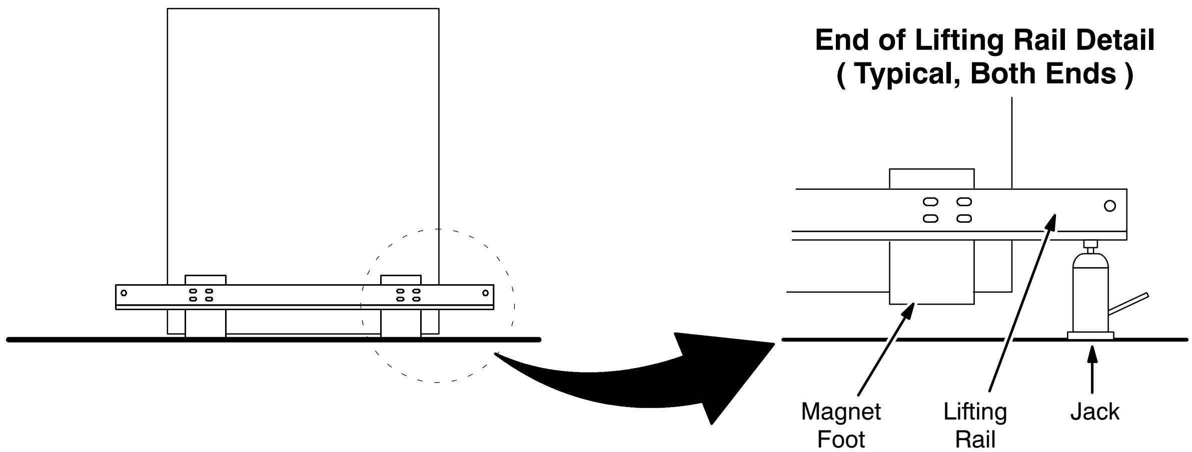
CAUTION

Keep the magnet level at all times. Uneven jacking of the magnet's corners could result in the magnet shifting on jacks.

3. Raise the magnet using jacks as in Illustration 6-5 and insert 4 x 6 inch (100 x 150 mm) wooden beams under the lifting rails at each end of the magnet and a roller dolly under the each end of the wooden beams. The centerline of each jack must be at least 2 inches (50 mm) from the end of each lifting rail and centered on the horizontal leg of the angle.

Note
The magnet is shipped with lifting rails installed. See Illustrations 6-2 and 6-4.

6-2 MOVING THE MAGNET (continued)



JACKING THE LIFTING RAILS
ILLUSTRATION 6-5

4. Compare the dimensions of the magnet on roller dolly with the clearances measured along the magnet delivery route. If the minimum overhead clearance will be less than the 100.3 inches (2617 mm) the magnet normally requires, the 1 inch (25 mm) spacer plate can be removed from the magnet feet. The height can also be reduced by putting the roller dollies directly under the lifting rails.

Note

Adjustments to magnet / roller dolly height to meet clearance needs are best made before moving the magnet. If simple adjustments are not sufficient to meet those needs, call the MAC Team representative for removal of plumbing / shim lead components.

6-2 MOVING THE MAGNET (continued)

CAUTION

Do not apply any loads to any enclosure cover parts nor allow straps / cables / chains to scrape enclosure cover parts.

5. Move the magnet to the magnet room. If using a motorized tow vehicle, attach cables, chains or straps to the magnet's lifting rails with shackles.

Note

If there are turns in the delivery route, adjust the roller dollies to appropriate positions to negotiate turns.

Use shims to move a magnet on roller dollies over door thresholds or other inclines.

If the actual overhead clearance encountered while moving the magnet is less than the 100.3 inches (2617 mm) the magnet normally requires, the 1 inch (25 mm) spacer plate can be removed from the magnet feet. The height can also be reduced by putting the roller dollies directly under the lifting rails. If further adjustments are required to meet overhead clearance needs, call the MAC Team representative for removal of plumbing / shim lead components.

6. Identify the table end of the magnet before moving into magnet room.
7. Position the magnet over the anchor studs.

Note

It is recommended to protect anchor stud threads with tubing to prevent damage to the threads during magnet installation.

CAUTION

Keep the magnet level at all times. Uneven jacking of the magnet's corners could result in the magnet shifting on jacks.

8. Jack the magnet up sufficiently at the 2 lifting rails (4 corners) and remove the roller dollies and wooden beams.
9. Slowly lower the magnet onto the magnet room floor. Release pressure simultaneously in both jacks on one end of the magnet until that end is 1 – 2 inches (25 – 50 mm) lower than the opposite end. Then simultaneously lower both jacks on the other end 1 – 2 inches (25 – 50 mm). Repeat lowering the magnet end to end until all magnet feet are on the floor, correctly located on the anchor studs.